The ID layer

The technical specification of how a verified human proves themselves online: the key hierarchy, credential formats, BBS selective disclosure, the Groth16 presence proof over the trace log, wrapped passport and EUDI attributes, the OpenID4VP flow, revocation, and the threat each mechanism addresses.

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Also the rig · the hardware spec · the command chain · the P0 bench

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IN ONE PARAGRAPH

Proofs travel, identity stays home.

The ID layer is the protocol stack that takes Vault's continuously verified human online without disclosing identity. The hub's secure element signs a per-second checkpoint into the trace log while the checks hold; BBS credentials carry attributes with selective disclosure and unlinkable derived proofs; a Groth16 circuit proves a fresh checkpoint sits in the log without revealing which; wrapped passport and EUDI proofs carry legal attributes; OpenID4VP 1.0 transports the presentation. This page specifies the key hierarchy, the credential formats, the proof constructions, the presentation flow, revocation, and the threat each mechanism addresses, then a build plan with pass marks.

Standards are cited as of October 2026, and drafts are labelled as drafts. "Proposal" marks design choices made for Vault, and the pass marks below are proposals, not established benchmarks.

01 · THE IDEA

What only Vault can prove

Deployed zero-knowledge identity systems verify once, at enrollment. zkPassport scans a passport chip — more than 130 countries — and generates the proof on the phone, so the verifier gets an answer, never the document[1]. Anon Aadhaar verifies the Indian government's RSA signature over the Aadhaar QR code inside a circuit[2]. World ID's Orb checks iris uniqueness once, for about 18 million people, and stores a Proof of Human on the phone — a proof of human, explicitly not of identity[3, 4, 5].

None of them can prove the sentence online services need: the verified owner's body is at the keyboard as of thirty seconds ago. The rig produces that all day the rig; this layer carries it online.

SYSTEM	PROVES	CHECKED WHEN	THE GAP FOR ONLINE ID
zkPassport	The document is genuine and its attributes hold	Once, at the scan	Says nothing about who holds the phone now
Anon Aadhaar	The holder possesses a government-signed ID	Once, at enrollment	Same, and the project is still marked under development[2]
World ID	A unique human exists	Once, at the Orb	Uniqueness without continuity or identity
Vault	A specific, living, continuously verified human is present	Continuously, all day	What this page specifies

02 · KEYS AND ANCHORS

Keys and trust anchors

KEY	HELD BY	ALGORITHM	SIGNS
Hub root key	The hub's secure element (SE051)	P-256 or Ed25519	Checkpoints; certifies every other hub key
BBS issuer key	Hub software, certified by the root key	BLS12-381	Z1 attribute credentials
Wallet key	The phone's hardware keystore (Secure Enclave / StrongBox class)	P-256	Key binding on every presentation
Attribute-source keys	Passport document signers, EUDI member states, banks	RSA or ECDSA (external)	Z3 source documents and credentials

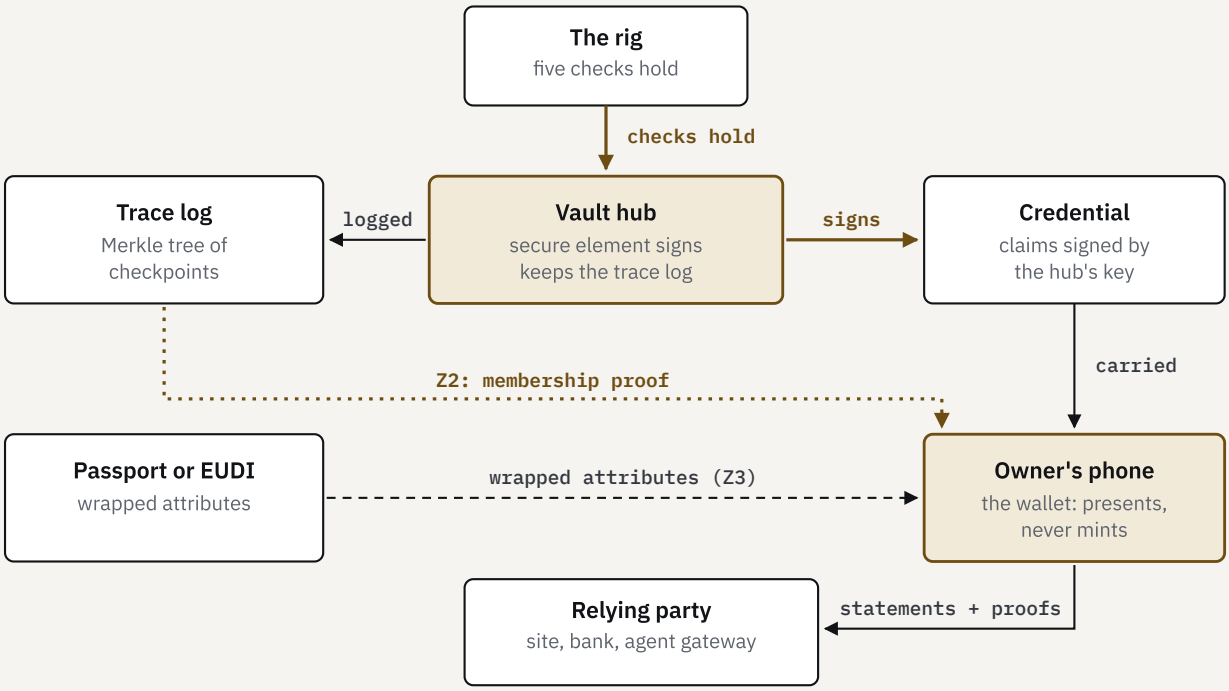
KEY	HELD BY	ALGORITHM	SIGNS
Verifier trust	Relying-party trust lists, eIDAS-style	—	Which issuer keys a site accepts

The SE051 signs P-256 and Ed25519 but not pairing-based BLS12-381[6], so the BBS issuer key is a hub software key certified by the secure element at pairing; checkpoint signing never leaves the element. BBS keys and proofs are defined over BLS12-381[7].

03 · CREDENTIALS

The credentials

The hub issues the credential in the issuer-holder-verifier shape of the Verifiable Credential model[8]. Four kinds exist:



How identity travels (proposal). The hub signs credentials only while the checks hold, and the phone presents and never mints, holding presenting rights through the command chain's narrowing tokens. The Z2 proof (dotted) shows a hub-signed checkpoint sits in the trace log's Merkle tree without revealing which, and wrapped attributes arrive as proofs generated on the phone[1, 10].

CREDENTIAL	CLAIMS	LIFETIME	SIGNED BY
Checkpoint (trace-log leaf)	Epoch timestamp, checks-held bitmap, owner commitment	60 s of validity	Hub root key, in the secure element
Presence proof (Z2)	None — a proof about a checkpoint	One presentation	The wallet, as a Groth16 proof
Attribute credential (Z1)	adult, resident-EU, per-verifier pseudonym, wallet-key commitment	Minutes to hours; long-lived attribute form up to 90 days	BBS issuer key
External credential (Z3)	Age, nationality, PID claims	Per source	The external issuer

Predicates are issuer-signed booleans, not range proofs: at enrollment the hub reads the wrapped passport's date of birth, computes "adult", and signs the boolean (proposal). The holder discloses true or false; no birthdate is ever signed into circulation.

04 · Z1 IN DETAIL

Selective disclosure, in detail

Issuance. The hub BBS-signs the claim set plus a commitment to the wallet key, producing a constant-size signature of about 80 bytes[9, 10, 13].

Presentation. The wallet runs ProofGen over the signature, disclosing only the chosen claims, and produces a derived proof of about 272 + 32 × (number of hidden claims) bytes[13]. The proof is zero-knowledge over the hidden claims and the signature itself, and is unlinkable: the issuer cannot connect it to issuance, and two sites cannot connect it to each other[7, 10]. The presentation header carries the verifier's nonce and audience, binding the proof to one request[7].

Key binding. The wallet key is a committed message inside the credential; each presentation also carries a wallet-key signature over (nonce, audience), the same replay defense as SD-JWT's KB-JWT[11]. A credential lent to another phone fails this check.

Per-verifier pseudonym. The wallet derives λ_v = HMAC(wallet_secret, verifier_id) and includes it as a claim, disclosed on request: the same site sees the same pseudonym, different sites see unrelated ones (proposal).

Format choice. BBS is the default because it is the only signature algorithm being standardized that natively supports selective disclosure[13]. SD-JWT[11, 12] is the fallback where a JWT-only verifier exists, accepting its weaker unlinkability[22].

05 · Z2 IN DETAIL

The presence proof, in detail

What is proven. A checkpoint signed by the hub root key sits in the trace log's Merkle tree, and its timestamp is within the last 60 seconds. Nothing about which checkpoint, which owner, or which checks leaks.

The log. Leaves are the per-second checkpoints; the tree is Poseidon-hashed (SNARK-friendly) at depth about 25 for a year of leaves. The hub publishes roots to a transparency log witnessed by owners' phones, the structure the report already uses for the trace log; verifiers fetch the current root set. A single-hub tree hides the checkpoint but not the hub; cross-owner anonymity needs a shared log across hubs, the Certificate Transparency pattern (proposal, listed in section 11).

The circuit (Groth16 over BN254 or BLS12-381[14]):

PUBLIC INPUTS	PRIVATE INPUTS
Merkle root R, hub root public key, freshness bound t_min, verifier nonce, nullifier N_v	Checkpoint leaf, its signature, Merkle path, wallet secret s

Constraints: the signature verifies under the public key; the path connects the leaf to R; the leaf timestamp ≥ t_min; and N_v = Poseidon(s, verifier_id). The nullifier is the Semaphore pattern: one site sees a stable pseudonym, two sites cannot link[15]. Proof size is about 192 bytes and verification is two pairing checks, a few milliseconds server-side[14]. Phone-side proving for circuits of this size is the class zkPassport already runs[1]; the pass mark below sets the time budget.

06 · Z3 IN DETAIL

Wrapped legal attributes

Passports. The chip implements ICAO 9303 passive authentication: the document signer's key, chained to the country root, signs the data groups[16]. The phone reads the chip, and the circuit verifies that signature, checks the age predicate against the date-of-birth group, and emits a proof plus a nullifier — zkPassport's deployed construction[1]. Nothing else in the chip leaves the phone.

EUDI wallet. The phone receives member-state credentials over OpenID4VCI and presents them under the EU's ZKP profile[17, 18]. Vault adds the Z2 presence proof alongside; it does not replace the wallet. The obligation for every member state lands in December 2026, and 24 of 27 will miss it, so interop is real but staggered[19].

07 · PRESENTATIONS

The presentation protocol

01 The relying party sends a signed OpenID4VP 1.0 authorization request: a DCQL query for the claims it may ask for ("adult", "presence within 60 s"), its nonce and audience[20].

02 The wallet checks the relying party's registration on a trust list, the same mutual identification the EU model requires[18].

03 The wallet derives the Z1 proof and runs the Z2 prover against the request nonce, then adds the key-binding signature.

04 The relying party verifies: the issuer trust anchor; BBS ProofVerify or Groth16 verification; key binding over (nonce, audience); the freshness window; revocation state. It logs a hash of the presentation, never the presentation itself.

05 Consequential presentations route through Vault's own gateway and confirm like any consequential command; a duress-marked credential is suppressed rather than presented (proposal).

Certified OpenID4VP verifiers have existed since August 2026, with more than 30 jurisdictions on the stack[21].

08 · REVOCATION

Revocation and expiry

Presence needs no revocation: the credential dies in 60 seconds, and a hub that stops seeing its owner simply stops signing. Attribute credentials live up to 90 days, so they need it: the issuer publishes a cryptographic accumulator over revoked serials, and the presentation proves non-revocation in-circuit, keeping unlinkability[7]. Status-list bitstrings are the fallback, at the cost of some cross-site correlation[22]. Succession follows the report's heir ceremony: a new hub root re-issues.

09 · THREAT MODEL

Threat model

ATTACK	STOPPED BY
Replay of a captured presentation	Key binding over (nonce, audience); the presentation header[7, 11]
Credential sharing (lending the phone)	Key binding with the hardware wallet key, plus Z2 freshness: presence requires the rig, which is on the owner
Cross-site correlation	BBS unlinkability and per-verifier nullifiers[7, 15]; the EU flags this as the part weaker formats miss[22]
Forged attributes	BBS signature; Groth16 verification over the hub signature[7, 14]
Stale presence	The 60 s freshness bound in the circuit
Stolen credential	The 60 s TTL; the hub stops signing when checks fail
Coerced presentation	Duress suppression; residual risk documented in section 11
Long-term quantum exposure	BBS and Groth16 are pairing-based; circuits stay portable to hash-based proving (proposal)

10 · BUILD PLAN

Build plan

PHASE	BUILD	PASS MARK (PROPOSAL)
Z0, phone only	A wallet holding a BBS credential from a test issuer[9]; a demo relying party speaking OpenID4VP 1.0[20, 21]; per-verifier pseudonyms on the Semaphore pattern[15]	A Z1 presentation generates in under 1 s and verifies in under 100 ms; two sites cannot correlate 100 presentations
Z1, with the P1 hub	The secure element signs checkpoints; the Groth16 circuit[14] proves membership and freshness; presentations flow through the gateway	The Z2 proof generates in under 10 s on a 2026 phone (under 2 s as the optimization target) and verifies in under 50 ms; zero biometric-template or log-content bytes transmitted, asserted from captures; a duress-marked credential is suppressed in at least 99 of 100 trials

PHASE	BUILD	PASS MARK (PROPOSAL)
Z2, interop	zkPassport-style wrapping; EUDI presentations against the ETSI ZKP profile[18]	One demo site accepts a Vault-native proof and an EUDI credential through the same flow

Z0 is pure software and runs in parallel with the P0 bench.

11 · OPEN PROBLEMS

Open problems

- **Attribute anchoring.** Vault self-certifies presence; "born 1990" needs the passport, a bank or the EUDI wallet as source. All three are wrapped, none is stored.
- **Cross-owner anonymity.** A single-hub log hides the checkpoint, not the hub; hiding the owner across a population needs the shared transparency log of section 5, with its own governance.
- **Coercion.** Every tier passes for a coerced owner. Duress suppression covers presentations, but a verifier that watches the owner type can infer; the duress design must cover the online path explicitly.
- **Quantum.** Pairing-based throughout today; the swap path to hash-based proving is a port, not a redesign, only if the circuits stay small.
- **The SE boundary.** BBS lives in hub software because pairing curves are not in the SE051[6]; a pairing-capable secure element or an HSM changes that line of the key table.

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